CSE 291 class projects

Spring 2026

The goal of the class project is to provide a foundation for learning how the principles taught in class apply to the analysis of real-world problems. In brief, we will do this by analyzing real world datasets in two stages:

- Stage 1: find associations between the data vectors and labels for the selected dataset. These associations can be determined in various ways, including by applying machine learning (ML) techniques that you may have used in the past or would choose to use now. Below we describe a basic way to determine associations using one-dimensional tests for differences of means/medians, but further challenge you to propose/investigate better ways to detect associations. Possible ways to do this include methods that detect more associations at the same level of significance or methods that can detect joint associations of multiple variables to the condition of interest.

- Stage 2: critically evaluate the quality of the associations detected in stage 1. This is the stage of the project that is most relevant for the purpose of the class, as it will require applying concepts taught in class to confirm by multiple lines of evidence that the detected associations are correct and not just spurious artifacts of the association methods. The multiple steps to be considered for this stage are described below in the corresponding section.

To facilitate getting started on the projects, we also provide a "Jupyter notebook and starter code" tutorial to illustrate how to load the data and do some simple analysis steps using a Jupyter notebook.

Stage 1: Datasets and project types

The datasets made available for class projects were selected to enable the analysis of important open questions with direct practical implications in real world problems, as described in the following subsections for each type of project. The longer description of each dataset is provided at the end of this document.

Condition classifier projects

Condition classifier projects aim to define models for associations between data vectors and class labels. This is the kind of project that is most similar to traditional machine learning problems, especially in its simplest form where there are only a small number of possible classes, such as the following dataset

- MSV000085507, COVID19 sera: 4 classes, 90 patients (1 sample per patient)
    - The groups in this dataset cover cases where patients (i) are healthy, (ii) are symptomatic without COVID19, (iii) have non-severe COVID19 or (iv) have severe-COVID19. The original publication focused on the last two groups (non-severe vs severe COVID19) but reanalyses can consider any combination of the 4 groups.

In addition, the following dataset can be used to define a variation of the classification problem:

- MSV000080596, Plasma weight loss: 58 patients, 7 time points per patient, total 336 samples
    - The number of classes for this dataset depends on the discretization of study outcomes. For example, patients could be divided into two groups based on whether the weight increased or decreased at the end of the study. Alternatively, patients could be divided into groups based on similarity of the weight progression time series throughout the course of the study.

Sample identifiability projects

Sample identifiability projects aim to define models to determine whether pairs (or generally, sets) of samples come from the same source. A very important instance of this type of question is to determine whether two samples come from the same patient, but it also extends to determining whether samples come from the same cell line or from different tissues of the same individual (e.g., human or model organisms).

- MSV000080596, Plasma weight loss: 58 patients, 7 time points per patient, total 336 samples
    - Samples acquired at different type points from the same patient should be detected as coming from the same source, while avoiding false positive matches between samples from different patients.
    - All of these samples should also be detected as coming from different patients than those in the dataset “MSV000085507, COVID19 sera”, where the data was also derived from blood samples but from different patients in a different country.

In addition to the sample-matching problem, it is also possible to determine the worldwide identifiability of a sample in cases where the sample contains evidence for enough mutations (caused by genetic variants) to reduce the chance of matching to a randomly-selected individual to less than 1 in 10 billion (which is slightly higher than the number of humans on Earth). This can be determined using the worldwide frequencies of occurrence of each mutation, which are available as additional data for this type of project.

Drug targets project

The drug targets project aims to determine the sets of proteins whose abundances are significantly perturbed by treatment with a drug with at least a given concentration. This is important because detecting which proteins become over- or under-expressed in response to a drug treatment indicates that the drug could be useful to treat conditions where the protein expression changes in the opposite direction of the drug treatment (e.g., a drug that lowers the abundance of protein P may be useful to treat conditions where P is over-expressed).

- MSV000081932 Target Landscape of Clinical Kinase Inhibitors: 284 drugs, 8 concentrations each + 284 control runs (one per drug treatment experiment)
    - Control runs for all drug treatment experiments can be used to determine the ``background’’ range of variation in protein abundances measured as a result of biological variation and of the experimental procedures used for acquisition of mass spectrometry data. This background distribution can then be used to find changes in protein abundance (in drug treatment experiments) that have less than 5% chance of being explained by the background variability of experimental measurements (e.g., a p-value of less than 0.025 at the end of each tail of the background distribution).

Spectrum matching projects

Spectrum identification requires either direct matching of a query spectrum to a set of sequences (spectrum/sequence matching) or matching to a set of identified spectra (spectrum/spectrum matching). Spectrum matching projects will explore different scoring functions for these matching questions for both (i) selecting the best identification per query spectrum and (ii) maximizing the number of total identifications at a fixed false discovery rate. These projects will use subsets of results from dataset MSV000083508, 29 tissues proteome map.

- Scoring Peptide-Spectrum Matches (PSMs), used to rank possible peptide sequences against each query spectrum
    - Training set: Lung tissue
        - Set of 184,621,938 top-20 PSMs per spectrum (2.7GB zip, 44GB uncompressed)
            - Use top Target/Decoy PSM per spectrum (across all spectra) to get training instances of true/false positive PSMs. Decoy PSMs (false positives) are those mapped only to proteins with identifiers starting with the string “XXX_”
            - Use the provided top 20 PSMs per spectrum to train for selection of the best PSM per spectrum
        - Set of 1,135,588 PSMs (and top-score ties) currently identified at 1% FDR (link)
    - Evaluation of results: Colon tissue
        - Set of 169,709,236 top-20 PSMs per spectrum (2.5GB zip, 41GB uncompressed), use to
            - Re-rank spectrum/sequence matches per spectrum
            - Determine set of spectra identified at 1% FDR (using Target/Decoy matches)
        - Set of 904,588 PSMs (and top-score ties) identified at 1% FDR (link)
            - Use for comparison of set-level spectrum identification results at 1% FDR

- Scoring Spectrum-Spectrum Matches (SSMs), used to score spectrum-spectrum matches to evaluate whether two spectra were generated from the same peptide
    - Training set: Duodenum tissue
        - Set of 846,462 PSMs identified at 1% FDR for 324,626 variants (link)
            - Full set of 1,783,205 spectra (4.2GB zip, 8.7GB uncompressed)
            - True-positives: any pair of spectra with the same “Variant group”
            - False-positives: any pair of spectra with the same charge state and with matching parent mass (e.g., within 0.05 m/z tolerance) but identified to different variants
    - Evaluation of results: match query spectra from the Colon tissue to the MassIVE-KB reference knowledge base (spectral library) and use decoy matches to obtain search results at 1% FDR
        - Colon tissue
            - Full set of 1,077,399 spectra (1.2GB zip, 3.1GB uncompressed)
            - Set of 73,552 PSMs identified by spectral matching at 1% FDR (link); use for comparison of set-level spectrum identification results at 1% FDR
        - MassIVE-KB spectral library, 4,281,960 spectra (2GB zip, 5.8GB uncompressed)
            - ~2.14M target spectra, where the PROTEIN field is not a decoy
            - ~2.14M decoy spectra, where the PROTEIN field
                - Starts with “DECOY_” - decoy entries generated by the shuffling method
                - Has only “XXX_” protein identifiers - decoy entries retained by the KB construction algorithm
            - Peptide/Variant sequence is reported in the “SEQ” field for each spectrum in the MassIVE-KB file

Stage 2: Critical assessment of detected associations

Once mass spectrometry identifications are estimated to be significantly associated with conditions of interest, it will be necessary to investigate the quality of the underlying identifications to determine whether these can be confirmed or need to be corrected, or even removed from the set of significant discoveries. This will be done by considering the following validation criteria:

- Peptide identification should be established by critical evaluation of the Peptide Spectrum Matches (PSMs) available for the peptide. Good matches should be supported by high-quality PSMs (e.g., high percentage of explained intensity) and preferably also by good matches to reference MassIVE-KB spectra or other search results PSMs from the same or related peptides.
  [This topic will be discussed in class during in the first half of the quarter and should thus be considered first]

- Modified variant identification should meet all criteria for peptide identification and should also meet additional criteria.
  - First, these should be supported by matching PSMs for the modified and unmodified variants of the same peptide. While in some cases it may be possible to find a reference spectrum for the modified variant in MassIVE-KB, it will be much more likely to find a reference spectrum for the unmodified version of the same peptide. The search results for the same dataset are also very likely to contain PSMs for the unmodified version of the same peptide.
  - Second, it is important to determine whether the detected mass offset (i.e., putative modification mass) is precisely localized to the indicated amino acid or more ambiguously localized to a region of the spectrum corresponding to two or more amino acids. In addition, it is also important to verify whether the detected mass offset has only one (or conversely, more than one) possible interpretation listed in the UniMod knowledge base of known modification masses.

  [This topic will be discussed in class during the last weeks of of the quarter and should thus be considered at that time]

- Disease and class association will be considered when ranking the importance of peptides to consider for inspection and rigorous validation. In particular, peptides found to have the highest association with the dataset classes, drug concentrations or sample identifiability, and thus reported as candidate markers for these detections, will need to be evaluated according to the criteria defined here.
  [Analyses for this topic depend on whether the associations are with peptides, proteins or modified variants, and should thus be done/updated when the corresponding topics are covered in class]

- Protein identification should be confirmed by high quality peptide identifications (i) that map uniquely to the protein of interest or (ii) that do not map uniquely to just one protein but are assigned to a TopProtein selected by the protein identification algorithm. Confidence in a protein identification is also increased by the consistent detection of the same peptides across all/most samples where the protein is identified.
  [This topic will be discussed in class during the middle of the quarter and should thus be considered at that time]

- Differential abundance of peptides and proteins will assess the significance of the degree of change in the abundance of peptides and proteins across conditions of interest. One example of such a statement would be the claim that a protein P increases in abundance by 2-fold in the disease group in comparison to the control group, and to report that such an observation could happen by random chance with a p-value of 0.01 (which is lower/better than the typical threshold of 0.05).

- While different metrics of association may be considered in stage 1, it will be important to estimate and report at least the Student t-test p-value for the comparison of the means of the distributions in the groups of interest (assumes Gaussian distributions), with a slightly better approach being to report the non-parametric test for comparison of medians (Wilcoxon ranksum test, does not assume Gaussian distributions).
- In addition, it will also be important to consider whether the peptides/proteins found to have significant change across conditions (i) do or (ii) do-not have consistent results for all samples in each group. For example, a peptide P may have significantly different means in the disease/control groups but there may still be some samples in the disease group with P at about the same abundance that is typically observed in the control group. In such cases, it will be important to consider whether other peptides from the same/different proteins may help to better separate the groups using *sets* of peptides instead of single peptides to separate the groups of interest.

[These metrics of association are standard statistical techniques that will not be directly covered in class but can be computed using standard approaches/tools]

- <u>Attribution of differential abundance</u> (DA) will consider the question of whether the important level of differential abundance is at the (i) protein, (ii) peptide sequence or (iii) modified variant.
  - <u>Protein-level DA</u> should be reported when all/most peptides assigned to the protein are detected to be differentially abundant. One way to assess this would be to test whether the peptide-level fold-changes are the same/different across the groups of interest. Another possible way to assess this is to check whether most peptides assigned to the protein (and covering enough samples in each group) are individually reported as DA.
  - <u>Peptide-level DA</u> should be considered in cases where protein-level DA is inconsistent (e.g., not confirmed by a majority of peptides covering the majority of samples in each group). In such cases, it is better to report differential abundance at the level of the peptides with significant DA associations, while also reporting the results for other peptides for the same protein without significant DA associations. In particular, it is important to consider whether the results may differ depending on whether the peptides do or do-not map uniquely to a single protein.
  - <u>Variant-level DA</u> should be considered in cases where (i) the unmodified peptide with the same sequence is not reported as DA and (ii) most other modified variants of the same sequence are not reported as DA. In these cases it is possible that the variant modification (which could be mutation/sequence polymorphism) could be the important factor to associate with a disease/group of interest. This type of association is also the kind that is likely to be most relevant when considering the identifiability of samples, which tends to be primarily determined by the detection of sequence variations that are detected in mass spectrometry by delta masses corresponding to sequence variations (i.e., polymorphisms).

  [Analyses for this topic depend on whether the associations are with peptides, proteins or modified variants, and should thus be done/updated when the corresponding topics are covered in class]

Project progression and evaluation

As mentioned in the previous section, different topics required for the class projects will be covered in class at different times during the quarter. It is thus expected that the main focus of the project will evolve throughout the quarter, with the initial work being on the study design and approaches for stage 1, and with work later in the project focusing on the inspection/validation steps required for stage 2.

Important dates to consider:

- Project plan due: April 20th
- Preliminary results due: May 13th
- Final results due: June 1st
- Project presentations: June 2nd, 4th and 9th (attendance required)

Evaluation guidelines:

- As a general guideline, it is more important for the analysis to be rigorous than to report outstanding results that do not stand up to scrutiny.
- Projects will be evaluated using criteria similar to those used to evaluate research papers
    - Design of the project plan; should rigorously define the proposed approach, with a clear definition of inputs/outputs and criteria for accurately measuring the possible degrees of success (e.g., precision/recall). The plan should also consider the possible limitations of the proposed approach and should mention possible alternative approaches to the data analysis in case the main approach does not work as expected.
        - See also Project Plans rubric for more details.
    - Description of project experiments; should clearly describe the steps taken to implement the project plan, including descriptions of adjustments to the plan found to be necessary in response to intermediate results.
        - See also Preliminary Results rubric for more details
    - Description of project results; should clearly describe the processes used to determine the most significant examples selected for evaluation, and should describe how such examples did or did not meet the evaluation criteria described for stage 2. In particular, claims of significance for specific peptides/proteins should be contrasted with the baseline statistical significance assessments of differential abundance mentioned in the "Differential abundance" section of stage 2.
        - See also Final results rubric for more details
    - Presentation of project results; given that there will be limited time to present the project results in class, it will be important to present results clearly and concisely with keen focus on the most relevant results and examples. That said, it is also important to be prepared to provide additional information about details of the project so it is strongly recommended to include supplementary materials (i.e., additional slides) describing the details of important parts of the project.
- We recommend that office hours be used to discuss the ideas, process and progress of the class projects as these develop throughout the quarter. This will make it easier to converge towards a well designed project that is possible to implement and validate in the available time frame, and will help avoid situations where design flaws detected late in the quarter would make it difficult to re-do all the analysis in time to meet the course deadlines.

Data types

The data provided for the class projects is made available in one of the following data types described below. All tables can be downloaded at the links provided for each dataset in "Dataset descriptions" using the same steps: (i) Click the "Download" tab at the top of the page, (ii) Set "Include Entries" to the "All" option and (iii) Click the "Download" button to get a zip file containing the results table.

- Variants intensities
  - Peptide identification information in these columns
    - “Variant”: identified sequence including modifications
    - “Unmod variant”: identified sequence without modifications
    - “Charge”: charge of the identified peptide
    - “Top canonical protein”: protein assigned to the peptide identification, with proteins separated by “;”
    - “Canonical proteins”: all proteins containing the identified peptide sequence
  - Peptide abundances are reported in pairs of columns with headers (X, X_unmod), where X corresponds to the group/class label
    - Projects should use values in the columns whose header does *not* include the “_unmod” suffix
    - Replace zeros with N/As (if needed)
    - The number of samples per group corresponds to the number of non-N/A entries per group
    - Variant abundances are proportional to their ion counts in the sample and can thus be compared/contrasted between different peptides (e.g., to determine modified/unmodified ratios per peptide sequence)
- Variants TMT
  - Peptide identification information in these columns
    - “Peptide”: identified sequence including modifications
    - “Unmodified_sequence”: identified sequence without modifications
    - “Charge”: charge of the identified peptide
    - “Top_canonical_protein”: protein assigned to the peptide identification, with proteins separated by “;”
    - “Canonical_proteins”: all proteins containing the identified peptide sequence
  - Peptide abundances are reported in columns with suffix “_intensity_for_peptide_variant”
    - Replace zeros with N/As (if needed)
    - The number of samples per group corresponds to the number of non-N/A entries per group
    - Variants TMT abundances indicate fold-variation across samples and are not directly comparable between different peptides
- Variants amino acid coordinates per protein - same as variant intensities, but with the additional relevant columns
  - “Start_AA”: starting amino acid (AA) position of the peptide on the indicated protein sequence
  - “End_AA”: end amino acid (AA) position of the peptide on the indicated protein sequence
  - Both “Start_AA” and “End_AA” are zero-based coordinates
  - The protein coordinates of the sequences of variants and peptidoforms can be obtained from this table by matching the unmodified peptide sequences
- Peptidoforms intensities
  - “Peptidoform”: identified sequence including modifications
  - “Unmod peptidoform”: identified sequence without modifications
  - “Charge”: charge of the identified peptide
  - “Proteins”: all proteins containing the identified peptide sequence, with proteins separated by “;”. Protein identifiers with a “-” or starting with “tr|” can be ignored for determining whether a peptide has a unique match to a single canonical protein
  - “Annotation without position”: type of modification or single amino acid polymorphism (SAAP) explaining the detected mass offset

        - SAAPs are shows as entries of type “X->Z” where both X and Z are single-letter amino acid codes
        - Note that the following cases do not correspond to SAAPs: "Cys->Dha", "Gln->pyro-Glu", "Glu->pyro-Glu", "Trp->Kynurenin"
    - “PValue”: significance of the best alignment of the modified peptide spectrum to a spectrum of the unmodified peptide from the same sample. P-values are shown as -log10(probability) so higher values represent better matches.
    - Peptide abundances are reported in pairs of columns with headers (X, X_unmod), where X corresponds to the group/class label
        - Projects should use values in the columns whose header does *not* include the “_unmod” suffix
        - Replace zeros with N/As (if needed)
        - The number of samples per group corresponds to the number of non-N/A entries per group
        - Peptidoform abundances are proportional to their ion counts in the sample and can thus be compared/contrasted between different peptides (e.g., to determine modified/unmodified ratios per peptide sequence)
- Peptidoforms TMT
    - Same as “Peptidoform intensities” above, with the following exceptions
    - Peptide abundances are reported in pairs of columns with headers (X, X-Unmodified sequence), where X corresponds to the group/class label
        - Projects should use values in the columns whose header does *not* include the “-Unmodified sequence” suffix
        - Replace zeros with N/As (if needed)
        - The number of samples per group corresponds to the number of non-N/A entries per group
        - Peptidoform TMT abundances indicate fold-variation across samples and are not directly comparable between different peptidoforms
- Single amino acid polymorphisms (SAAP) frequencies file: [SAAP_frequencies_dbSNP_2021.tsv]
    - “Accession”: The UniProt protein identifier (same as in search results)
    - “Variation”: The amino acid variation of type “X->Z” where both X and Z are single-letter amino acid codes
    - “Location”: The (1-based) position of the variation on the given protein
    - “refSNP cluster ID number: Accession for a variant type assigned by dbSNP [more info]”
    - “AC”: Allele count for the indicated allele in the general population. An allele is a specific type of amino acid and the indicated position on the protein
    - “AN”: Total number of observations in the whole population where the protein was observed, including counts for the allele in the reference genome.
    - “Frequency(ratio)”: The allele frequency is calculated as the allele count divided by the total size of the population where the protein was observed

Datasets descriptions

Proteomic and Metabolomic Characterization of COVID-19 Patient Sera [MSV000085507]

Dataset overview
- 5,586,392 spectra, 645 files totalling 99.4 GB
- 4 classes, 90 patients (1 sample per patient)

Paper abstract: Early detection and effective treatment of severe COVID-19 patients remain major challenges. Here, we performed proteomic and metabolomic profiling of sera from 46 COVID-19 and 53 control individuals. We then trained a machine learning model using proteomic and metabolomic measurements from a training cohort of 18 non-severe and 13 severe patients. The model was validated using 10 independent patients, 7 of which were correctly classified. Targeted proteomics and metabolomics assays were employed to further validate this molecular classifier in a second test cohort of 19 COVID-19 patients, leading to 16 correct assignments. We identified molecular changes in the sera of COVID-19 patients compared to other groups implicating dysregulation of macrophage, platelet degranulation, complement system pathways, and massive metabolic suppression. This study revealed characteristic protein and metabolite changes in the sera of severe COVID-19 patients, which might be used in selection of potential blood biomarkers for severity evaluation. [link to online paper]

Main search results for class project
- Full search results
- Variants TMT, with 101,461 variants detected in at least one sample (used for classification and regression models)
  - ModDecode results for all resolved peptidoforms
- Peptidoforms TMT, with 21,444 resolved peptidoforms detected in at least one sample (used for identifiability models)
- Variants amino acid coordinates per protein

Plasma Proteome Profiling Reveals the Effects of Weight Loss [MSV000080596]

Dataset overview
- 26,350,740 spectra, 2,687 files totalling 931.06 GB GB
- 7 classes, 58 patients

Paper abstract: Sustained weight loss is the preferred intervention in a wide range of metabolic conditions, but the effects on an individual's health state remain ill-defined and controversial. Here, we investigate the plasma proteomes of a cohort of 43 obese individuals that had undergone eight weeks of weight loss (average of 12%) followed by a year of weight maintenance. Using plasma proteome profiling, a mass spectrometry-based approach, we measured 1294 plasma samples to examine dynamic changes of hundreds of human plasma proteins, including the entire apolipoprotein family. Longitudinal monitoring of the cohort revealed individual-specific protein levels and functional connected proteins with correlated regulation over time. The wide-ranging effect of losing weight on the plasma proteome was reflected in 93 significantly affected proteins. The adipocyte secreted SERPINF1 and the apolipoprotein APOF1 were most significantly down and up regulated (p<10**13), although their fold-change was modest (-16% and +37%), respectively. Overall, markers of inflammation went down in nearly all study participants (39 of 42), however, there were diverse patterns of inflammation marker responses, allowing individual-specific interpretation of the effects of weight loss. Plasma proteome profiling can be used broadly to evaluate and monitor intervention in metabolic diseases. [link to online paper]

Main search results for class project

- Full search results
- Variants intensity, with 55,012 variants detected in at least one sample (used for classification and regression models)
- Peptidoforms intensities, with 40,921 resolved peptidoforms detected in at least one sample (used for identifiability models)
  - All known and novel modifications decoded from this dataset
  - All ModDecode results (one job per patient)
- Variants amino acid coordinates per protein
- Study values for standard laboratory values (including patient weights)

Target Landscape of Clinical Kinase Inhibitors [MSV000081932]

Dataset overview

- 87,761,553 spectra, 6,986 files totalling 1.45 TB
- 2,821 classes, 7 subjects

Paper abstract: Kinase inhibitors are important cancer therapeutics. Polypharmacology is commonly observed, requiring thorough target deconvolution to understand drug mechanism of action. Using chemical proteomics, we analyzed the target spectrum of 243 clinically evaluated kinase drugs. The data revealed previously unknown targets for established drugs, offered a perspective on the "druggable" kinome, highlighted (non)kinase off-targets, and suggested potential therapeutic applications. Integration of phosphoproteomic data refined drug-affected pathways, identified response markers, and strengthened rationale for combination treatments. We exemplify translational value by discovering SIK2 (salt-inducible kinase 2) inhibitors that modulate cytokine production in primary cells, by identifying drugs against the lung cancer survival marker MELK (maternal embryonic leucine zipper kinase), and by repurposing cabozantinib to treat FLT3-ITD-positive acute myeloid leukemia. This resource, available via the ProteomicsDB database, should facilitate basic, clinical, and drug discovery research and aid clinical decision-making. [link to online paper]

Main search results for class project

- Search results for 50 drugs
- Variants intensity, with 83,706 variants detected in at least one sample (used for drug response models)

A deep proteome atlas of 29 healthy human tissues [MSV000083508]

Dataset overview

- 80,831,472 spectra, 1,934 files totalling 2.12 TB
- 30 classes, 40 subjects

Paper abstract: Genome-, transcriptome- and proteome-wide measurements provide insights into how biological systems are regulated. However, fundamental aspects relating to which human proteins exist, where they are expressed and in which quantities are not fully understood. Therefore, we generated a quantitative proteome and transcriptome abundance atlas of 29 paired healthy human tissues from the Human Protein

Atlas project representing human genes by 18,072 transcripts and 13,640 proteins including 37 without prior protein-level evidence. The analysis revealed that hundreds of proteins, particularly in testis, could not be detected even for highly expressed mRNAs, that few proteins show tissue-specific expression, that strong differences between mRNA and protein quantities within and across tissues exist and that protein expression is often more stable across tissues than that of transcripts. Only 238 of 9,848 amino acid variants found by exome sequencing could be confidently detected at the protein level showing that proteogenomics remains challenging, needs better computational methods and requires rigorous validation. Many uses of this resource can be envisaged including the study of gene/protein expression regulation and biomarker specificity evaluation. [link to online paper]

Main search results for class project

- Full search results
- Peptidoforms intensities, with 560,583 resolved peptidoforms detected in at least one sample (used for identifiability models)
    - All known and novel modifications decoded from this dataset
    - All ModDecode results (one job per tissue type)
- Variants amino acid coordinates per protein